

CMP-06 · COMPETENCY FRAMEWORKS

Cost and commercial competencies in depth

The chain from estimate to final account, and where competence actually fails along it



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PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

— Summary

Cost and commercial competence is not a set of separate skills but one chain — estimate, budget, commitment, accrual, actual, forecast, change, final account — in which each link must reconcile to the last. This document takes the cost and commercial competencies that appear across all three PCI credentials, sets out what distinguishes a practitioner from a professional at each link, and shows with a worked example how a single error at one link is amplified by the time it reaches the forecast.

— About this document

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Contents

1. Why the cluster is one chain and not six skills	4
2. Cost management: the accrual is the competency	5
3. Earned value: the competency is validity, not arithmetic	6
4. Forecasting: the method is the competency	6
5. Commercial control: the record is the competency	7
6. The finance-side members of the cluster	7
7. How this goes wrong	8
8. Worked example: how far one accrual error travels	8
9. Checklist	10
Related	10
Sources and standards	10
Status and version	11

Cost and commercial competencies in depth

In one paragraph. Cost and commercial competence is not a set of separate skills but one chain — estimate, budget, commitment, accrual, actual cost, forecast, change, final account — in which each link must reconcile to the last. This document takes the cost and commercial competencies that appear across all three PCI credentials, sets out what distinguishes a practitioner from a professional at each link, and shows with a worked example how a single error at one link is amplified by the time it reaches the forecast. After reading it you should be able to identify which link in your own function is weakest and what evidence would prove it.

Who this is for. Cost engineers, cost managers, quantity surveyors, commercial managers and project controls leads; project finance analysts whose models depend on a construction cost position; and anyone assessing competence claims in this cluster.

— 1. Why the cluster is one chain and not six skills

Cost and commercial work is organised in most businesses as separate roles: estimating, cost control, commercial management, planning, finance. The competencies are assessed separately too. But the numbers they produce are sequential, and each one inherits every error made upstream of it:

estimate → budget → commitment → accrual → actual cost → forecast → change → final account

A person can be flawless at their own link and still be part of a function that reports a wrong number, because the failure is at a handoff. In practice the handoffs are where competence separates, and they are where assessment should look:

Handoff	The question competence answers	The failure when it is absent
Estimate → budget	What was in the estimate that is not in the budget, and who agreed to absorb it?	Budget set below the estimate to fit an approval, with no record of the gap
Budget → commitment	Is the committed value the whole obligation, including options and provisional items?	Commitments understated; the project appears to have contingency it has already spent
Commitment → accrual	What work has been done that has not been invoiced?	Accrual driven by invoices received rather than by progress achieved
Accrual → actual	Does the project cost report reconcile to the ledger, and are the differences explained?	Two cost positions in one organisation, neither owned
Actual → forecast	Which method extrapolates this performance, and what must be true for it to hold?	The tool's default formula, applied to unvalidated inputs
Forecast → change	Is agreed change in the baseline, and unagreed exposure disclosed outside it?	Change buried in the forecast, so the movement cannot be explained
Change → final account	Does the record substantiate the entitlement to someone who was not there?	A defensible entitlement lost for want of a contemporaneous record

The competencies in this cluster, across the three credentials, are: cost management, earned value management, forecasting and EAC, performance measurement and commercial and contract controls (PCL-AI); investment appraisal, financial modelling, capital structure, debt and equity, project contracts and commercial and financial risk (PFL-AI); business cases, procurement, contracts and commercial delivery and cost, schedule and risk integration (PML-AI). They are defined and levelled in CMP-03, CMP-04 and CMP-05. This document goes underneath the definitions.

— 2. Cost management: the accrual is the competency

Most cost control work is mechanical and can be learnt in a quarter. One judgement in it cannot, and it is the accrual: the value of work done that has not yet been invoiced.

A practitioner raises accruals from a defined process — usually a supplier statement, a goods-received record or a progress claim. A professional decides the accrual where those inputs disagree with physical reality, and can defend the number to a finance function that would prefer a smaller one at year end. The specific judgements are:

- **Work done off site.** Fabrication in progress at a supplier's works represents cost incurred and, in most contracts, no payment obligation yet. Whether it is accrued, and on what evidence, is a decision.
- **Disputed valuations.** A contractor claims 400 units of progress; the project assesses 320. The accrual is not the average, and it is not zero because it is disputed. It is the project's honest assessment, with the difference disclosed as exposure.
- **Rework.** Cost incurred on work that will have to be done again is a real cost and a fictional progress. The two entries are separate, and conflating them is how a project reports on budget until the month the rework is discovered.
- **Cut-off.** The same rule, applied on the same date, every period. A month with a late cut-off flatters itself and steals from the next.

The evidence that distinguishes the levels is not the accrual value; it is the substantiation. A signed accrual pack in which each line names its basis — progress record, supplier statement, physical verification, assessed value with the difference disclosed — is professional-level evidence. A spreadsheet of numbers is not, whatever the numbers are.

— 3. Earned value: the competency is validity, not arithmetic

The indices are arithmetic — see BPG-08 — Earned value in practice for the method. The competency question is whether the earned value is true.

Three tests separate the levels, and an assessor can apply all three in ten minutes.

Does the rule of credit reward work or activity? A 0/100 rule on a six-month work package produces five months of nothing and a month of everything, and hides the trend the system exists to reveal. A percentage-complete rule based on the manager's opinion produces steady 8 % per month regardless of what happened. A professional selects the technique per work type — units complete, milestone weighting, level of effort where nothing else is honest — and can say why for each.

Has anyone compared claimed progress to physical progress? The single most valuable half-day in cost control is walking the work with the progress claim in hand. A candidate who has never done this has been operating a system, not validating one.

What happens to level-of-effort scope? Scope measured by time passing always earns exactly as planned and therefore always shows perfect performance. When it is a large share of the baseline, the project's indices are structurally optimistic. A professional knows their level-of-effort share and states it when presenting the indices.

— 4. Forecasting: the method is the competency

An estimate at completion (EAC) is a claim about the future, and every forecasting method encodes an assumption about how the past predicts it:

- **Performance-index extrapolation** ($EAC = BAC \div CPI$, equivalently $BAC \times AC \div EV$) assumes remaining work will be performed at the efficiency achieved so far. Defensible when the remaining work resembles the completed work and the drivers of underperformance persist.
- **Remaining-work re-estimate** ($EAC = AC + \text{a bottom-up estimate of what is left}$) assumes the past performance does not predict the future — typically because the cause was one-off, or because the remaining work is different in kind. Defensible when you can say what changed.
- **Composite approaches** apply different treatments to different scopes: index-based on the like-for-like remainder, re-estimated on the parts that have changed.

Competence is visible in three habits, all of which leave evidence:

1. **The method is chosen and recorded, not inherited.** A dated note naming the method, the alternatives rejected and the reason is professional-level evidence and takes ten minutes to write.
2. **The assumption is stated with the answer.** "EAC 4.4 million, assuming the remaining civils are performed at the current cost performance index of 0.909 and no further scope is instructed" is a forecast. "EAC 4.4 million" is a number.

3. **The forecast is back-tested.** A professional knows whether their own forecasts have historically run low, and by roughly how much. This is the cheapest credibility available to a controls professional and almost nobody does it.

BPG-09 — Estimate at completion owns method selection in full, and TPL-08 provides the scenario comparison sheet.

— 5. Commercial control: the record is the competency

Commercial competence is often assessed by contractual knowledge, which is the wrong test. Contract knowledge is available in the document. What is not available later is the record of what happened.

The professional-level behaviours, stated as artefacts:

- **A notice issued within its period.** Most contracts impose time limits on notification, and an entitlement that was real becomes unrecoverable when a date passes. The competency is a system that triggers the notice, not a memory that recalls the clause.
- **Contemporaneous records that a stranger can use.** Allocation sheets, delay records, photographs with dates and locations, minutes recording instructions. The test: could someone who was not on the project reconstruct entitlement from this file in eighteen months, when the people have left?
- **Exposure kept outside the forecast and visible.** An unagreed variation, a claim in preparation and a potential liability are exposure. Burying them in the forecast destroys the ability to explain a movement; omitting them entirely misleads. The professional reports the forecast and the exposure range separately, and says which is which.
- **A position that concedes what is not defensible.** Competence includes recognising a losing argument early and settling it, rather than carrying a fictional entitlement in the commercial report for a year.

BPG-11 — Change orders and variations and BPG-12 — Claims and extension of time own the mechanics.

— 6. The finance-side members of the cluster

For PFL-AI candidates the same chain runs through different artefacts. The construction cost position feeds the drawdown profile; the contract suite determines whether a cost overrun is borne by the project or by a contractor; and the commercial exposure that a controls function reports outside the forecast is precisely what a lender's technical adviser is looking for. Two boundaries matter for assessment:

- **Financial modelling competence is not cost management competence.** A faultless model implementing an unvalidated cost position produces confident, wrong cover ratios. The competency to interrogate the underlying cost position is a separate one and is the more valuable of the two.
- **Investment appraisal competence is a decision competence.** It is evidenced by the statement of what is being discounted, at what rate and whose return it represents, not by the presence of a net present value in a paper. See CMP-04 §2.4.

— 7. How this goes wrong

The budget is set below the estimate and the gap is not recorded. An approval is granted at a number the estimator did not support, the difference is absorbed silently, and eight months later the overrun is attributed to poor cost control. The competency is the recorded reconciliation between estimate and budget, produced at the time by someone who was willing to have the conversation.

Accruals are driven by invoices. The cost report then measures the supplier's administrative efficiency rather than the project's progress, and it looks best in the months when suppliers are slowest.

Two cost positions coexist. The project cost report and the general ledger differ, both are quoted in different meetings, and nobody owns the reconciliation. The competent response is a monthly reconciliation with each difference explained and aged — not a campaign to make the two systems identical, which will fail.

The forecast method changes without disclosure. A project switches from index-based to a re-estimate in the month the index-based number became unwelcome. The change may be entirely legitimate; not disclosing it is not. State the method every period, and flag the period in which it changed.

Exposure is either buried or omitted. Buried, it destroys the explanation of forecast movement. Omitted, it misleads. Separate reporting of forecast and exposure range is a professional-level habit and the fastest way for a candidate to demonstrate this competency.

Contemporaneous records exist but are unusable. Photographs without dates, allocation sheets without locations, minutes that record attendance rather than instructions. The record is only worth what a stranger can reconstruct from it.

Earned value is validated against the system that produced it. Checking that the report agrees with the tool confirms the export worked. Validation means comparing claimed progress to something outside the system — physical inspection, an independent count, a supplier's own records.

Cost competence is assessed on report quality. The report is the last link and the easiest to make attractive. Assess at the accrual, the reconciliation and the forecast note.

— 8. Worked example: how far one accrual error travels

Illustrative figures. A single control account, currency-neutral cost units (cu), status at month 8 of a 20-month scope. Monetary results rounded to the nearest 1,000 cu; indices shown to three decimal places but computed from unrounded values.

Reported position at month 8

Item	Value (cu)
Budget at completion (BAC)	4,000,000
Planned value to date (PV)	2,400,000
Earned value to date (EV)	2,200,000
Actual cost to date (AC)	2,420,000

Cost performance index: $CPI = EV \div AC = 2,200,000 \div 2,420,000 = 0.909$ Schedule performance index: $SPI = EV \div PV = 2,200,000 \div 2,400,000 = 0.917$ Cost variance: $CV = EV - AC = 2,200,000 - 2,420,000 = -220,000$

Performance-index forecast, computed without rounding by using the equivalent form $EAC = BAC \times AC \div EV$:

$EAC = 4,000,000 \times 2,420,000 \div 2,200,000 = 4,000,000 \times 1.100 = 4,400,000$ $VAC = BAC - EAC = 4,000,000 - 4,400,000 = -400,000$

The commercial layer. Three items sit outside the baseline: variations agreed at 150,000 but not yet incorporated; two instructed but unpriced variations assessed by the project at 90,000, against the contractor's submitted 140,000; and an intimated extension-of-time claim with no cost quantification.

Incorporating only the agreed variations gives a revised BAC of $4,000,000 + 150,000 = 4,150,000$. The forecast now depends on an assumption that must be stated:

- If the variation work is assumed to perform to plan: $EAC = 4,400,000 + 150,000 = 4,550,000$
- If it is assumed to perform at the current efficiency: $EAC = 4,150,000 \times 2,420,000 \div 2,200,000 = 4,150,000 \times 1.100 = 4,565,000$

The difference is $4,565,000 - 4,550,000 = 15,000$, which is simply $150,000 \times 0.100$. Small here, and not small when the variation value is a quarter of the budget. **Reporting either number without naming the assumption is the failure.**

Taking the second treatment as the reported forecast: $VAC = 4,150,000 - 4,565,000 = -415,000$

The unpriced variations are **not** in that number. They are disclosed as exposure of 90,000 to 140,000, unresolved, plus an unquantified time-related cost. Assumptions the answer depends on: no further scope instructed; the current efficiency persists across the remaining work; the disputed valuation is settled at the project's assessment.

Now the error. Suppose the month-8 accrual was understated by 60,000 — one supplier's off-site fabrication omitted because no invoice had arrived. The true actual cost is $2,420,000 + 60,000 = 2,480,000$.

$CPI = 2,200,000 \div 2,480,000 = 0.887$ $EAC = 4,150,000 \times 2,480,000 \div 2,200,000 = 4,150,000 \times 1.127 = 4,678,000$ (to the nearest 1,000 cu)

The forecast moves by $4,678,000 - 4,565,000 = 113,000$ — roughly **1.9 times** the size of the original error. That multiple is not a coincidence: an index-based forecast multiplies an error in actual cost by $\text{revised BAC} \div EV$, here $4,150,000 \div 2,200,000 = 1.886$, and $60,000 \times 1.886 = 113,000$ to the nearest 1,000 cu.

The point for competence assessment: the person who owns the accrual is usually junior, the accrual is usually the least reviewed number in the pack, and the forecast that the board argues about

for an hour inherits its error amplified. This is why cost management competence is assessed at the accrual and not at the report.

— 9. Checklist

Take this into a cost and commercial review, or into an assessment of evidence in this cluster.

- [] Is there a written reconciliation between the estimate and the approved budget, with the gap named?
- [] Are commitments complete, including options, provisional items and unpriced instructions?
- [] Does each accrual line name its basis, and can any line be traced to physical evidence?
- [] Is the cut-off date the same as last period, and is adherence recorded?
- [] Does the project cost report reconcile to the ledger, with differences explained and aged?
- [] What proportion of the baseline is level-of-effort scope, and is that stated when indices are shown?
- [] Has anyone compared claimed progress to physical progress this period?
- [] Does the forecast note state the method, the alternatives rejected and the assumptions?
- [] If the method changed this period, is the change disclosed?
- [] Are agreed changes in the baseline and unagreed exposure reported separately as a range?
- [] Have all notices due under the contract been issued within their periods?
- [] Could a stranger substantiate the current entitlement position from the record file alone?

— Related

- [CMP-03 – PCL-AI: the fourteen competencies](#) — the definitions and levels for the controls-side members of this cluster
- [CMP-04 – PFL-AI: the nineteen competencies](#) — the finance-side members
- [CMP-05 – PML-AI: the twenty-four competencies](#) — procurement, contracts and commercial delivery
- [BPG-09 – Estimate at completion – choosing and defending a method](#) — the forecasting methods in full
- [BPG-11 – Change orders and variations](#) — the change mechanics behind \$5
- [TPL-08 – Estimate at completion scenario comparison](#) — the working sheet for \$4

— Sources and standards

- PCI platform certification catalogue ([backend/Data/MultiCert.cs](#)), verified August 2026 — the seeded competency sets from which this cluster is drawn.
- PCI Body of Knowledge domains 3, 5, 6, 7 and 11 ([docs/bok/](#)) — budgeting and forecasting, cost management, earned value, contracts and commercial management, and business process cycles. Cited by domain; not reproduced.
- The earned value quantities and index definitions used in §7 are the standard published definitions used across the discipline and are described in our own words. No standard's text, tables

or worked examples are reproduced. All figures in §7 are illustrative and recomputed from first principles.

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